

Oldest junk

US satellite, the Vanguard I, which launched in 1958 and worked for only six years, enjoys the unsavoury distinction of being the oldest debris still on orbit



India's first satellite

Aryabhata, India's first satellite weighed 360 kg and was launched by the erstwhile Soviet Union on April 19, 1975

Cleaning up the Cosmos

We're talked about e-waste before, but there's a whole other type that's threatening our future – orbital debris, or space junk, as it's more popularly known...

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In September this year, Boeing announced its intention to purchase a ticket for the new space race. It's not all about getting spy satellites and ICBM busters into orbit anymore. The future is space tourism! If Boeing's efforts go as planned, by 2015, a lucky few of us will blast-off from Florida's Cape Canaveral in a state-of-the-art "Space Taxi". Of course, this Space Taxi won't be as crowded as the transportation we're used to...it will have a maximum of seven people onboard – at least three of which will be professional astronauts. And you won't even have to pay extra for meals, or water!

British tycoon, Richard Branson, truly believes that space is the final frontier. He founded a company called Virgin Galactic, which is expected to begin test flights beyond Earth's atmosphere in 2012.

In less than a decade, space around Earth is going to get crowded – teeming with tourists. However, it's already crowded up there. In fact, Mumbai's crowded local trains would seem empty, in relative terms.

Dodging bullets

We're surprised no one's calling their space project Neo, after Keanu Reeves' character in the Matrix trilogy. Why? Because launching into space these days is like dodging Agents who are shooting at you with sub-machine guns.

Future space tourists in so-called Space Taxis will actually face higher risks than drivers on the worst Indian roads.

On the periphery of our atmosphere, there are about 5,00,000 objects flying about at breakneck speed. Nuts, bolts, shards of metal, plastic, etc. – all remnants of previous space missions, and old satellites, and all deathly dangerous.

Now, before you accuse us of sensationalism, and think, "Sheesh, talk about over-hyped! Nuts and bolts indeed," try and imagine what a nut would feel like if it were shot at you at over 40,000+ kilometers per hour! At such speeds, a nut would go *through* a tank; and in space, that's one dead tank.

Already there have been numerous cases of near misses for vital satellites orbiting the earth. On one occasion NASA had to replace a window on a space

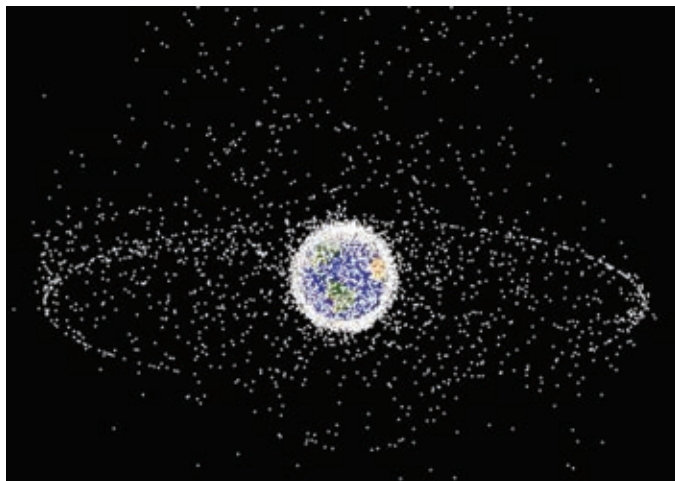
shuttle, and that damage was caused by a mere flake of paint – one tenth of a millimetre wide!

In February this year, the world woke up to the news that a near head-on collision between the residual portions of a spent Chinese rocket and the European Space Agency's (ESA) huge Envisat Earth remote-sensing spacecraft had been narrowly averted. The tracking information released by the US and Germany shows that these two objects had floated past each other at a nail-biting distance of a mere 160 feet. The ESA's Envisat tips the scales at 8 tonnes, and the old Chinese rocket is estimated to weigh 3.8 tonnes. In space, that's like two bullet trains at full speed missing each other by micro-meters.

What's worse, if such a collision had taken place, the threats of space junk would have increased in orders of magnitude because of all the additional debris that these two huge objects would have left behind after they were destroyed. Thankfully, scientists on the ground were able to steer the ESA Envisat out of harm's way. All we can say is we have new-found respect for the astronauts who live aboard the International Space Station – flirting with constant threats.

Monitoring the junk

Mankind is yet to come up with a workable solution to clearing our own garbage in space. With increased activity in space – DTh and HD channels need satellites now, don't they – the amount of junk is only going to rise. IN fact, if we don't find a



Images from NASA's orbital debris site might be able to emphasize the perils of space junk a lot better than all of the words in this article!